$$
\mathtt{A}^{lat} = \left[\begin{array}{cccc:ccc:c:cc:c}
\ell_{1,1} & \ell_{1,2} & \dots & \ell_{1,m} & e_{n+1} & \dots & e_{2n} & \dots & 0 & \dots & 0 \\
\vdots & \ddots & \ddots & \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\
\ell_{n,1} & \ell_{n,2} & \dots & \ell_{n,m} & 0 & \dots & 0 & \dots & e_{n^2+1} & \dots & e_{n^2+n} \\
\hdashline
0 & 0 & \dots & 0 & e_1 & \dots & 0 & \dots & 0 & \dots & 0 \\
\vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \dots & \vdots & \ddots & \vdots \\
0 & 0 & \dots & 0 & 0 & \dots & e_1 & \dots & 0 & \dots & 0 \\
\hdashline
\vdots & \ddots & \ddots & \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\
\hdashline
0 & 0 & \dots & 0 & 0 & \dots & 0 & \dots & e_n & \dots & 0 \\
\vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \dots & \vdots & \ddots & \vdots \\
0 & 0 & \dots & 0 & 0 & \dots & 0 & \dots & 0 & \dots & e_n
\end{array}\right], \tag{3}
$$